0580_w18_qp_33_Q8

8 **(a)** A bag contains 20 bulbs.
8 are yellow, 5 are red, 4 are white and 3 are pink.
Sam takes one bulb at random.

Find the probability that the bulb he takes is

(i) white,

..... [1]

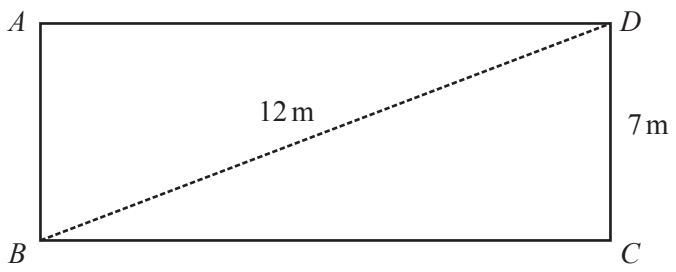
(ii) blue,

..... [1]

(iii) not pink.

..... [1]

(b) Sam has a rectangular pond, *ABCD*.



NOT TO SCALE

(i) Calculate *BC*.

BC = ... m [3]

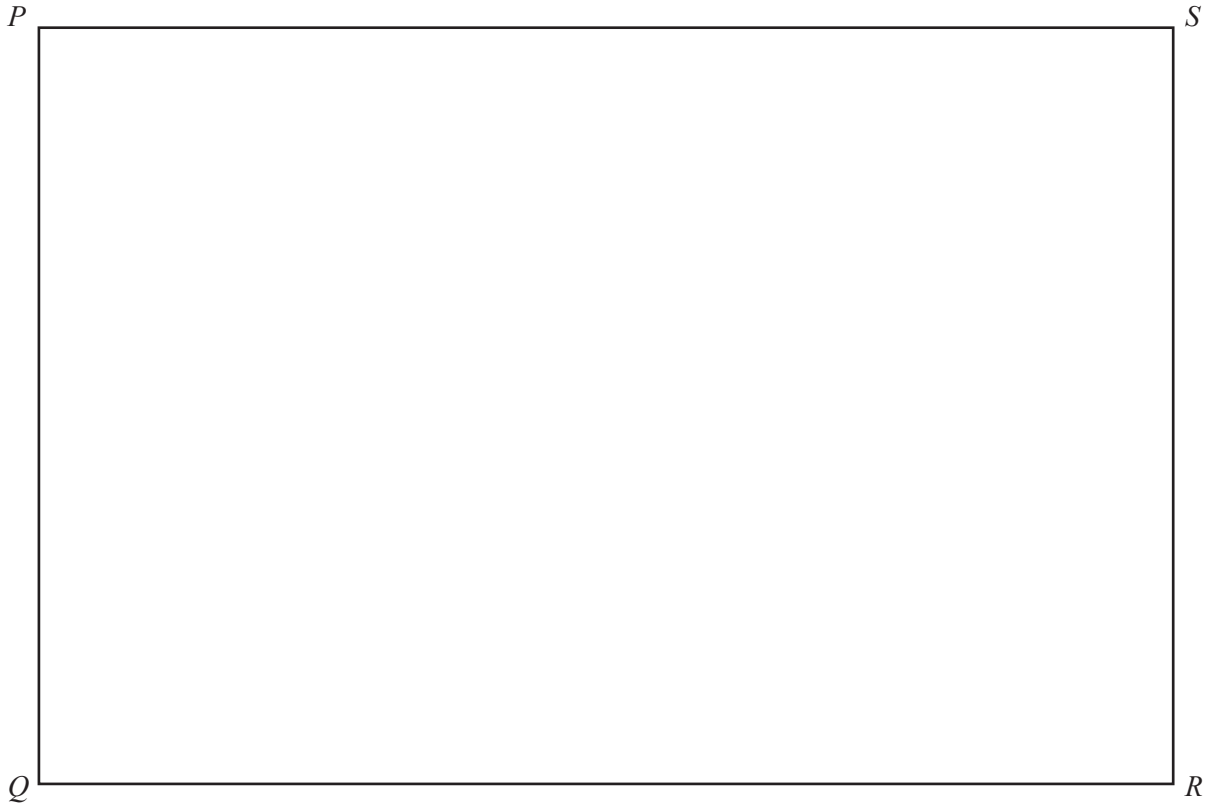
(ii) He puts a fence around the edge of the pond.

Calculate the length of the fence.

..... m [1]

0580_w18_qp_33_Q8

(c) A scale drawing of Sam's garden, *PQRS*, is shown below.
The scale is 1 centimetre represents 4 metres.



Scale : 1 cm to 4 m

Sam plants some bulbs so that they are

- less than 30 metres from *P*

and

- nearer to *PQ* than to *PS*.

Using a ruler and compasses only, construct and shade the region where he plants the bulbs. [5]